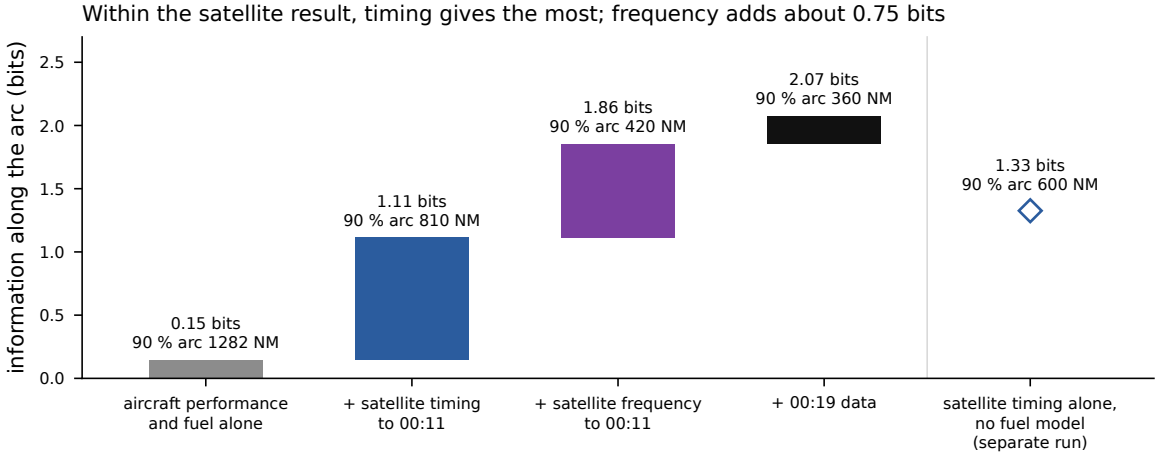
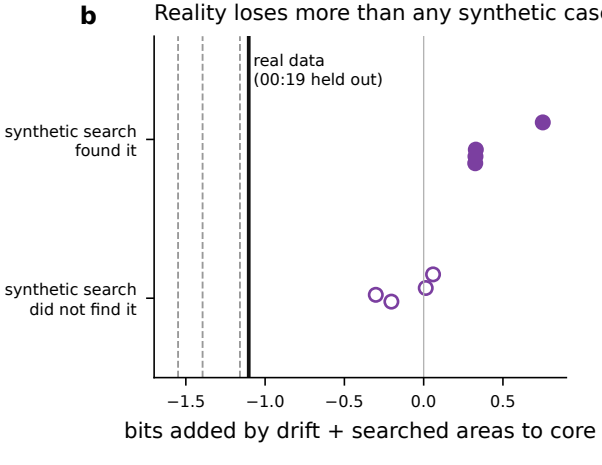
Next steps: what makes up the satellite result, and a check with synthetic cases

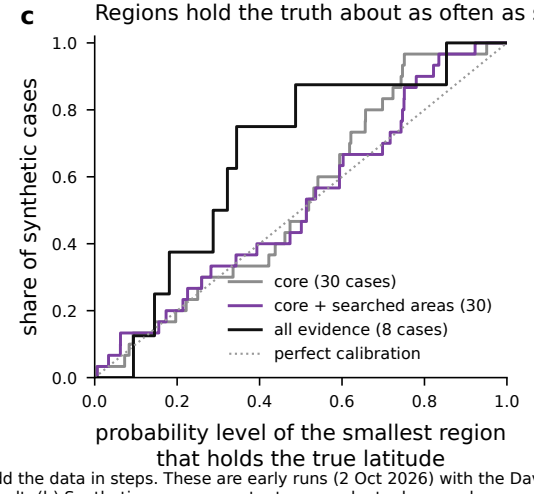
a



b



c



Note (STE100): (a) How much each part of the satellite result narrows the 7th arc. We add the data in steps. These are early runs (2 Oct 2026) with the Davey flight model and the early fuel model; they are not run J. So the numbers are a guide, not a result. (b) Synthetic cases: we put a true wreck at a known place, we made all the data from it, and we ran the full method. Each dot is one synthetic case. In these cases, drift and the searched areas change the result by a small amount. With the real data, they remove 1.1 bits. This agrees with the disagreement test (about 2 sigma). But there are only 8 cases from about 4 different flights, and no case is like the real one (the satellite peak on searched ground). (c) A check that the method does not claim too much: a line below the dotted line means the stated regions are too small. All lines are close to it or above it.

Technical: (a) runs endurance_1 to _4 (4 x 14 M, Davey priors, BFO 7 Hz, track 295.66 deg, tau 0.1-10 h, fuel flow factor N(1.0085, 0.0178), endurance proposal, no radar, no one-engine flight) and davey-bto-only-2026-10-18 (run J version of the Davey reproduction, BTO only, no fuel); 00:19 latitude PDFs on 0.05 deg mapped to 30 NM bins of the 00:19 arc (small circle 0.57 N 64.25 E, radius 44.50 deg), 40.25-19.9 S, renormalised; fuel alone = fuel-only 2-D map cells within 0.25 deg of the arc. Bits against uniform along the arc. (b) staged end-to-end calibration (20-21 Oct 2026, tier 1a' event, 4 x 3.5 M per case): H(core) - H(all) on 0.25 deg impact-latitude bins, D0, the 8 cases with drift scored by the data-only rule; real value from the evidence ledger (commit ead63fa) on the same bins; dashed: real values for 00:19 BTO only, both BFOs as recorded, and Holland bias. (c) HPD level of the true impact latitude (0.05 deg cells), D0; provisional, calibration at reduced depth (30 cases, 16 distinct truth flights).